Testing connectome-constrained Drosophila brain models with simulation-based inference: the limits of dopaminergic neuron loss and predictions for Parkinson's disease

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Abstract

Whole-brain connectomes of *Drosophila* now exist for both sexes, and leaky integrate-and-fire (LIF) models built on them predict sensorimotor circuits with high accuracy. Whether such models can also simulate dopamine-dependent phenomena, the basis of fly models of learning and of Parkinson's disease, has not been tested. We built a cell-type-resolved lesion framework on the FlyWire female connectome (138,639 neurons) and the MaleCNS male connectome (164,606 neurons), validated the simulator against the reference model (per-neuron rate correlation 0.963), and measured how progressive loss of dopaminergic clusters (PAM, PPL1, PPL2, PPM1) changes network output under a six-stimulus battery, in an embodied body model, and through an amortised inverse model that infers lesions from phenotypes. In the model's default regime, loss of the 16 PPL1 neurons shifts the phenotype six times more than loss of 16 random central neurons, through three labelled lines (PPL105→MBON13, PPL103→MBON31, PAM08→MBON29) that switch mushroom-body output neurons off one at a time. This specificity, however, is an artefact: every odour input we tested drives the model into a brain-wide attractor in which 75 % of Kenyon cells fire and output neurons saturate at 160 Hz. Restoring sparse Kenyon-cell activity (3.8 %) by strengthening APL inhibition abolishes every dopaminergic lesion effect: an inverse model trained on 1,000 random lesions of 32 dopaminergic subtypes recovers six subtypes in the attractor regime and none in the sparse regime (surrogate R^2 0.52 against −0.02), and progressive loss of whole clusters stays within the noise floor (PPL1 distance 1.74 against a noise floor of 1.91, random loss of 307 neurons 1.84), with or without a dopamine-as-modulator layer that we fitted to conditioning data, and at synaptic weights 27 % below and above the reference (at the stronger weight the effect returns only as the network re-approaches the attractor). The reason is structural-dynamical: natural reinforcers (sugar, bitter, water) and mechanosensory inputs never activate PAM or PPL1 neurons (0 Hz), although sugar gustatory neurons reach 69 PAM neurons within two synapses. The only drive to dopaminergic neurons in the model is the odour attractor. Tracing the pathways shows why: the neurons that sugar activates send at most 1.9 mV to any PAM-projecting relay, no strengthening of that route up to 16-fold makes PAM fire, and the octopaminergic neurons that carry sugar reward in vivo receive no sugar input and barely contact PAM, whereas bitter taste reaches PPL1 through 20 identifiable relay neurons that fire once their input and output synapses are strengthened 4-fold, a testable prediction of where the bottleneck lies. Re-running the neuron-loss protocol of a recent amyloid-pathology simulation (Jo, 2026) in both regimes reproduces its ordering (hub damage worst, antihub mildest) but shows that its whole-brain activity readout reverses sign when a single APL neuron is lost. Supplying the reward route as one extrasynaptic term turns the negative result into a test: every odour-responsive output neuron that behavioural work assigns to appetitive memory then learns from sugar, with a silent reserve that differs by compartment (learning intact after 50 % PAM loss, dependent on which neurons die at 75 %, gone at 90 %). The same model yields testable predictions for fly Parkinson models: the connectome's

synaptic load ranks PPL1 neurons 12-fold above PAM, matching the clusters lost in parkin and PINK1 flies; the order of loss sets when a memory fails while locomotor and passive readouts never show the loss; memories stored before the loss are kept while new ones cannot be formed; and tonic dopamine replacement restores nothing but, if the gate does not adapt, corrupts plasticity specifically in the compartments that learn. We conclude that connectome-constrained LIF models, as currently parameterised, can rank structural damage but cannot propagate reinforcement to dopaminergic neurons, and that one explicit extrasynaptic term is enough to make them a predictive tool for dopamine-dependent learning and its loss. All tools are released as open code.

Introduction

The complete synaptic wiring of the adult *Drosophila* brain is available for a female (FlyWire; Dorkenwald et al., 2024; Schlegel et al., 2024) and, since September 2026, for the whole central nervous system of a male (MaleCNS; Berg et al., 2026). Shiu et al. (2024) showed that a leaky integrate-and-fire network with synaptic signs assigned from predicted neurotransmitters, and otherwise no free parameters beyond a single synaptic weight, predicts which neurons respond to taste and which drive feeding and grooming, with 91 % agreement against activation and silencing experiments. The same architecture has since been run on the male connectome and used to simulate amyloid-like damage (Jo, 2026).

The fly is also the most widely used animal model of Parkinson's disease. In *parkin* and *PINK1* mutants, dopaminergic neurons of the PPL1 and PPM clusters degenerate with age and climbing declines (Whitworth et al., 2005; Yang et al., 2006; Wang et al., 2006); in α-synuclein models, loss of a subset of PAM neurons that project to the mushroom-body β′ lobe tracks the locomotor decline (Riemensperger et al., 2013). Dopaminergic neurons are also the teaching signal of associative memory: reward and punishment reach PAM and PPL1 neurons, which depress the Kenyon-cell-to-output-neuron synapses of their compartments (Aso et al., 2014; Hige et al., 2015; Owald et al., 2015).

A connectome-based model that reproduced these phenomena would allow cell-type-resolved, progressive lesions to be related to behaviour, and the inverse question, which lesion explains an observed phenotype, to be posed computationally. We asked whether the Shiu model supports this. We built the lesion and readout machinery, an amortised inverse model, a dopamine-as-modulator extension, an embodied readout in NeuroMechFly, and a male-connectome pipeline, and we report that the model's dopaminergic signal is an artefact of an odour-driven attractor and that reinforcement never reaches dopaminergic neurons. The negative result is the main finding; the tools are the secondary contribution.

Results

Simulator validation and lesion framework

Our NumPy re-implementation of the reference LIF model reproduces the published Brian2 output: on the reference protocol (21 sugar gustatory neurons, 200 Hz, FlyWire 630 tables) the per-neuron rate correlation over the 448 reference-active neurons is 0.963, 320 of the 333 neurons firing above 5 Hz are recovered, and 7 neurons are active in ours but silent in the reference (all below 10 Hz). Only neurons away from rest are updated each step, which makes a 1 s whole-brain run take about 10 s on one core.

Lesions are specified at cell-type resolution as a fraction lost per group, drawn in a fixed random order so that a 40 % lesion is a superset of a 20 % lesion. Three mechanisms are available (neuron removal, output silencing, synaptic scaling). The phenotype is a 654-dimensional vector: the mean log firing rate of 109 readout populations (descending-neuron types and output neuron types) under six stimuli chosen to keep the model in a stable regime (sugar, water and bitter gustatory neurons; one odour glomerulus, ORN_DM1; wind/gravity and head-

bristle mechanosensory neurons). Trial-to-trial noise between intact repeats sets the floor at an L2 distance of 1.68.

Progressive dopaminergic lesions act through labelled lines in the default regime

In the model's default regime, losing the 16 PPL1 neurons moves the phenotype 6.19 L2 units from the intact brain, six times the effect of losing 16 random central neurons (0.97) and far above the noise floor (1.68). Losing all 307 PAM neurons gives 3.00, against 1.77 for 307 random neurons. PPL2 (8 neurons) and PPM1 (16) have no detectable effect (1.04 and 0.87). Five lesion orders and two simulation seeds per point were used.

The PPL1 curve is not a graded reserve. At 60 % loss its value ranges from 1.96 to 6.28 across lesion orders, and the full-loss value is reached as soon as particular neurons are gone. Tracing the connectivity explains why: each affected output neuron receives almost all of its dopaminergic input from a single subtype, and the response is all-or-none.

Dopaminergic subtype	Output neuron	Direct input (mV)	Next strongest DAN input (mV)
PPL105	MBON13	78.6	1.4
PPL103	MBON31	60.8	0.8
PAM08	MBON29	56.4	11.6
PPL104	MBON28	20.1	1.9

A sweep over 240 lesions of 32 dopaminergic subtypes confirmed this from the data side: a ridge surrogate (five-fold cross-validated R^2 0.33) ranks PPL103 (sensitivity 10.5), PPL105 (4.8) and PAM08 (4.6) first, and the remaining 29 subtypes at noise level (1.3 to 2.3). A neural posterior estimator trained on the same sweep recovers only the PPL103 lesion fraction from a phenotype (r 0.98); for every other subtype the posterior collapses to the sparse prior.

The odour response is a brain-wide attractor, and a sparse regime abolishes the dopaminergic effects

Every odour input tested drives the model into the same persistent state (Figure 9A). Driving the 68 ORN_DM1 neurons at 40, 75 or 150 Hz gives 9,950 to 10,100 active neurons, 75 % of Kenyon cells firing and a mean output-neuron rate of 161 Hz; eight ORNs or a 50 ms pulse are enough, the state outlasts the input, and six other glomeruli (DL5, DC2, VC3, DM5, VA2, DA2) behave identically. Scaling every synapse by 0.6 still leaves 51 % of Kenyon cells active, and removing all 293,762 Kenyon-cell-to-Kenyon-cell synapses leaves 67 %, so the attractor is not mushroom-body recurrence. Multiplying the two APL neurons' inhibitory synapses onto Kenyon cells by 5 restores sparse coding (3.8 % of Kenyon cells, output neurons at 10 Hz, 20 of 96 output types above 5 Hz) while 5,900 neurons elsewhere remain active. The manipulation leaves the gustatory validation untouched: with APL inhibition scaled fivefold, the reference protocol still gives a rate correlation of 0.963 and recovers 320 of 333 responding neurons.

The attractor is not a property of our simulator. Running the same 68 ORN DM1 neurons at 150 Hz for 500 ms through the reference model's own unmodified Brian2 code (run_trial of Shiu et al., 2024) gives 64 % of Kenyon cells above 5 Hz (mean 33 Hz), 67 % of MBONs (mean 85 Hz), 81 % of PPL1 (113 Hz), 39 % of PAM (21 Hz) and 7,946 active neurons, against 75 %, 161 Hz, 206 Hz, 45 Hz and 9,945 in ours; rates are 30 to 50 % lower in the original, the regime is the same.

In this sparse regime (noise floor 1.91; 5 lesion orders x 2 simulation seeds) the dopaminergic lesion effects disappear. Full PPL1 loss gives 1.74 (SD 0.20) and full PAM loss 1.12 (SD 0.22), both below the floor, while

307 random central neurons give 1.84 (SD 0.15) and 16 give 1.16. The six-fold specificity of the default regime was therefore a product of output-neuron saturation: a saturated output neuron can only be switched off, and its dedicated dopaminergic subtype is the switch.

The result does not depend on how sparsity is produced. Raising the spike threshold of Kenyon cells instead of strengthening APL needs a shift of 32 mV (4.5 times the resting-to-threshold distance) before the odour-evoked KC fraction falls to 3.4 %, a measure of how strongly the attractor is wired; at that setting the network sits at the same operating point as APL ×5 (5,664 active neurons, MBONs 9.8 Hz, PPL1 13.9 Hz), the noise floor is 1.67, and complete loss of PPL1, PAM or 16 random neurons moves the phenotype by 1.22, 1.15 and 0.88. Both findings also survive a change of the inherited synaptic weight. At w = 0.20, 0.275 (reference) and 0.35 mV the odour drives 64, 75 and 79 % of Kenyon cells with APL at its native strength, and 2.6, 3.7 and 4.6 % with APL strengthened 5-fold. In the sparse regime the noise floor is 1.24, 1.55 and 6.66, and complete PPL1 loss moves the phenotype by 0.56, 0.96 and 9.61 (random 16: 0.63, 0.83, 3.94; one lesion order, two intact seeds). At the strongest weight the sparse network sits near the attractor boundary (7,489 active neurons against 5,674 at reference, MBONs at 16 Hz), its floor rises four-fold, and PPL1 loss becomes visible above it: the dopaminergic effect returns with the attractor, not independently of it.

Treating dopamine as a modulator removes the switch but not the problem

In the reference model dopaminergic synapses are ordinary fast excitatory synapses, which is what makes a dopaminergic neuron an on/off switch for its output neuron. We replaced this with the biological arrangement: dopamine released in a compartment sets the gain of the Kenyon-cell synapses onto that compartment's output neuron, and depresses them when it rises above its tonic level. Compartments were derived from the wiring (any dopaminergic neuron with at least 2 mV of direct input onto an output neuron), the fast dopaminergic synapses were removed, and the acute gain was taken relative to the tonic rate measured in the intact brain under the same stimulus.

We fitted the two free parameters by simulated conditioning, pairing odour with direct activation of one compartment's dopaminergic neurons. The chosen setting (gain 0.003 per Hz, plasticity 1e-4) depressed the paired output neuron MBON11 by 59 % with 16 % off-target change. The appetitive pair reached only 19 %, because its output neuron sits near the refractory ceiling in the attractor regime. The fit was made in the attractor regime and not repeated in the sparse regime; the sparse-regime results below use the same parameters.

With this layer the all-or-none behaviour disappears. Lesion curves become graded and far less dependent on which neurons are lost first: the PPL1 spread across lesion orders at 60 % loss narrows from 4.3 to 0.35, a twelvefold reduction. The direction of the effect also inverts, with output neurons disinhibited rather than silenced. But the total effect shrinks, and in the sparse regime the modulated model is no more readable than the reference one.

Group, 100 % loss	Reference, attractor	Modulator, attractor	Reference, sparse	Modulator, sparse
PPL1	6.19	2.23	1.74	0.89
PAM	3.00	4.00	1.12	1.12

Noise floors 1.68 (attractor) and 1.91 (sparse). Making dopamine a modulator fixes the mechanism but does not make dopaminergic loss visible to a passive sensory battery.

Reinforcers never reach dopaminergic neurons

The reason the model cannot report dopaminergic loss is that its dopaminergic neurons are almost never driven. We measured the mean firing rate of each cluster under every stimulus of the battery, in both regimes.

Stimulus	PAM, attractor	PPL1, attractor	PAM, sparse	PPL1, sparse
Sugar (reward)	0.0	0.0	0.0	0.0
Bitter (punishment)	0.0	0.0	0.0	0.0
Water	0.0	0.0	0.0	0.0
Odour, ORN_DM1	53.0	206.4	0.6	16.3
Wind and gravity	0.0	0.0	0.0	0.0
Head bristles	0.0	0.0	0.0	0.0

Rates in Hz, mean over the cluster, 100 to 500 ms after stimulus onset. Not a single PAM or PPL1 neuron fires under any reinforcer, at any synaptic scale we tested, including the unscaled model and drive at 300 Hz.

This is not a wiring gap. On the connectivity graph, 69 PAM neurons lie two synapses downstream of the sugar gustatory neurons and 225 lie three synapses away. The pathway exists but does not conduct: sugar activates a local suboesophageal population of about 418 neurons, and the relay neurons onto PAM never reach threshold. Sugar reward reaching PAM neurons is one of the best-established facts about this circuit (Liu et al., 2012; Burke et al., 2012), and the model does not reproduce it.

The consequence is that the only drive to dopaminergic neurons in this model is the odour attractor, which is itself an artefact. It also makes the obvious rescue experiment impossible: an associative conditioning protocol with a natural reinforcer cannot be run, because the unconditioned stimulus never reaches the teaching neurons. Driving the dopaminergic neurons directly, as we did for the parameter fit, measures the injected signal back out and tests nothing about the circuit.

Where the reinforcer pathway fails, and the smallest change that restores it

We traced the route from gustatory neurons to the dopaminergic clusters in the sparse regime. Sugar gustatory neurons (32) contact 414 neurons directly; 121 of them fire, mostly subesophageal interneurons at 85 to 200 Hz. This active population sends no appreciable input to any PAM-projecting neuron: the strongest two-synapse relay receives 0.3 to 1.9 mV (1 to 7 synapses) from the gustatory set. Raising the gustatory rate does not help: at 600 Hz the dopaminergic clusters stay silent, and at 1000 Hz, or with the 57-neuron sugar-plus-low-salt set at any rate, the brain enters the attractor and PPL2 and PPL1 fire as part of it. Water never reaches a dopaminergic neuron. Bitter gustatory neurons (42) activate 113 neurons, and these do drive PPL1-projecting relays (CB0135, SMP026, DNpe053, LAL154, SMP154), but at 2 to 4 mV per 10 ms, below threshold.

We then asked for the smallest local change that conducts. Taking the 20 relays with the largest product of input from the active set and output onto the target cluster, we scaled the input hop, the output hop or both by a factor k and measured the target cluster at a physiological 150 Hz reinforcer rate.

Reinforcer to target	Hop scaled	k	Target rate (Hz)	Neurons active	Side effects
bitter to PPL1	input	4	17.3	5,272 (attractor)	PPL2 42 Hz
bitter to PPL1	input	8	26.3	760	PPL2 11 Hz
bitter to PPL1	both	4	33.6 (50 % of PPL1 above 5 Hz)	822	PPL2 12 Hz; odour-evoked KC fraction unchanged
bitter to PPL1	output	16	0	113	none
sugar to PAM	both	16	5.4 (18 % above 5 Hz)	469	none

The punishment pathway exists in the wiring and conducts once 20 relay neurons are strengthened 4-fold on both sides; the model therefore makes a specific prediction about which neurons form the bottleneck. The reward pathway does not: no two-synapse route from sugar to PAM conducts at 16-fold strengthening. In vivo, sugar reward reaches PAM through octopaminergic neurons (OA-VUMa2 and the OAMB receptor; Burke et al., 2012), so we looked at the 43 octopaminergic neurons of the connectome directly. None receives a synapse from a sugar gustatory neuron, the network that sugar activates inhibits most of them (rate-weighted input of −0.3 to −7.5 mV per 10 ms) and none fires under sugar; 35 of the 43 make no synapse onto PAM, and the strongest (OA-VUMa6, OA-VPM3) send 3.3 to 3.6 mV in total onto 12 to 13 PAM neurons. Driving all 43 at 150 Hz gives PAM 0.5 Hz (3 % above 5 Hz) while PPL1 reaches 18 Hz. The reward route is therefore not weak in this wiring diagram; it is absent from it, consistent with octopamine acting on PAM extrasynaptically, which no synaptic connectome records.

With the punishment pathway restored (both hops 4-fold), we ran the conditioning readout that the unmodified model could not support: odour (ORN DM1, 150 Hz) paired for 1 s with bitter taste (150 Hz), dopamine-gated depression on, learning index = (pre − post) / pre of each MBON's odour response. During pairing PPL1 fired at 36 Hz and PPL106, the dopaminergic neuron of the α3 compartment, at 14 Hz; with PPL1 removed both were silent. The α3 output neuron MBON14 lost its odour response after pairing (index 1.0) in the intact brain and after loss of 16 random central neurons, kept it after loss of 50 % of PPL1 when PPL106 survived (seed 0) and lost it when PPL106 was in the removed half (seed 1), and showed no learning after complete PPL1 loss. The pattern is the expected one, but its magnitude is not: MBON14 responds to this odour at 0.8 Hz in the sparse regime, so the index measures 0.8 Hz going to 0. Other MBONs changed by less than 10 %. A quantitative learning curve would need a stronger Kenyon-cell drive onto MBON14, which is a further departure from the single-weight model and was not attempted.

A positive control that bypasses the pathway shows what the modulation layer can do once a teaching signal arrives. Following the training step of Jo (2026), we paired the odour with direct 100 Hz drive of the surviving PAM neurons and measured the learning index under progressive PAM loss and loss of 307 random neurons (two seeds each). MBON09, which receives 52 PAM neurons of four types (PAM12, PAM13, PAM14, PAM08; 244 mV in total), responds to the odour at 12.5 Hz and is silenced by pairing (index 1.0); that depression is unchanged after 25, 50 and 75 % PAM loss and gone (0.06) only at 100 %, while random loss leaves it at 1.0 throughout. MBON33 (index 0.55 intact) falls to 0.21 at complete PAM loss and MBON29 (3 Hz) degrades from 50 % loss. The learning curve therefore has a silent reserve: because the dopamine gate depends on the summed rate of the compartment's surviving dopaminergic neurons, a quarter of them driven at 100 Hz still opens it. The threshold lies where fly models of Parkinson's disease place the onset of symptoms, but here the

reserve sits in the convergence of dopaminergic neurons onto each MBON, not in the reinforcer pathway, which this protocol bypasses.

What the wiring predicts once a teaching signal arrives

Supplying the two missing routes turns the negative result into a test. We added the reward route as a single extrasynaptic term, the mean rate of the sugar gustatory neurons times 0.5 added to the effective dopamine rate of every surviving PAM neuron (the octopaminergic signal the connectome lacks), kept the 4-fold relay route for bitter taste, and paired the odour with each reinforcer for 1 s in the dopamine-gated model. Which MBONs learn is then decided by the wiring and the dynamics, not by us, and can be compared with the compartments that behavioural experiments assign to appetitive and aversive memory (Aso et al., 2014; Aso and Rubin, 2016). We ran the protocol at two operating points, the sparse regime (APL ×5, 7 MBON types respond to the odour) and an intermediate one (APL ×3, 20 % of Kenyon cells active, 12 MBON types respond), and measured the learning index against its unpaired noise (the same tests with odour alone, 3 seeds).

Operating point	MBON	Literature compartment	Memory assigned in the literature	Learning index (sugar)	Unpaired noise (max)
APL ×5	MBON09	γ3β’1	appetitive	0.88	0.12
APL ×5	MBON33	unassigned	unassigned	0.47	0.13
APL ×3	MBON07	α1	appetitive	1.00	0.03
APL ×3	MBON02	β2β’2a	appetitive	0.89	0.06
APL ×3	MBON12	γ2α’1	aversive	0.67	0.19

Every MBON that the literature assigns to appetitive memory and that responds to this odour learns from sugar in the model; the odour-responsive MBONs it does not assign (10, 20, 21, 26) do not. Of the assignments this comparison rests on, α1, β’2a and γ2α’1 were established by training flies with those compartments’ own dopaminergic neurons (Aso and Rubin, 2016), and the numbering of MBONs and dopaminergic types follows Aso et al. (2014) and Li et al. (2020); γ3β’1 rests on the reward role of the PAM cluster rather than on a compartment-specific experiment. One assignment is contradicted: MBON12, an aversive compartment, is also depressed by sugar pairing, but that depression survives complete PPL1 loss and random loss of 307 neurons, so it is not carried by its dopaminergic input and we count it as a model artefact. Loss of PAM neurons removes each appetitive memory with a silent reserve whose size differs by compartment: MBON07 still learns fully after 90 % PAM loss and fails only at 100 %, MBON02 fails at 90 %, MBON09 and MBON33 between 50 and 90 % depending on which neurons are lost (two orders here, five in the Parkinson section below); PPL1 loss and random loss change none of them. The reserve is not set by the number of converging PAM neurons alone (52 for MBON09, 20 for MBON07) but by the product of the Kenyon-cell drive the MBON receives and the dopamine excess that survives the loss.

The punishment arm fails for a reason that is itself a prediction. Bitter taste alone drives PPL101 at 46 Hz through the strengthened relays, but bitter together with the odour drives it at 0 Hz, and during odour-plus-bitter pairing the PPL1 cluster fires at the same rate as under odour alone (36 against 33 Hz): the odour-activated network silences the punishment signal. Decomposing the input onto PPL101 shows which synapse does it: under the odour PPL101 receives 133 mV per 10 ms of excitation (81 from Kenyon cells) and 241 of inhibition, of which 166 comes from the two APL neurons, the GABAergic cells whose inhibition of Kenyon cells we strengthened to obtain the sparse regime and which fire strongly under any odour; MBON feedback contributes 8. Under bitter alone APL is silent, the inhibition is 78 (55 from MBON11) and PPL101 fires at 46 Hz. Removing the APL-to-PPL1 synapses, and nothing else, makes PPL101 fire at 66 Hz under odour plus bitter (45 Hz under odour alone); removing the next two inhibitory sources (CRE102, SMP115) changes nothing.

With that one change, in the intermediate regime where the aversive compartments respond to the odour, bitter pairing depresses MBON12 (literature γ2α’1, an aversive compartment) by 0.33, the depression is abolished by complete PPL1 loss (−0.07) and absent after random loss (0.13), against an unpaired noise of 0.19: the model’s first PPL1-dependent punishment memory, in the right compartment, weak. MBON11 and MBON14, whose odour responses are saturated (125 and 61 Hz), still do not learn. The wiring-level claim is therefore specific: in this model the APL-to-PPL1 synapses close the punishment gate whenever an odour is present, and imaging PPL1-γ1pedc during odour-shock pairing with APL silenced is the test.

Predictions for fly models of Parkinson’s disease

With the reward route supplied, the model can be asked the questions fly Parkinson models ask, and it answers with specific, testable claims (Figure 8).

Which dopaminergic neurons should die first. The connectome itself ranks the clusters by synaptic load, the usual proxy for axonal arbor size and energetic demand: PPL1 neurons carry a median of 6,012 contacts each (PPL101 12,967), PPL2 4,236, PAM 481. Mitochondrial models are reported to lose PPL1 neurons: parkin mutants (Whitworth et al., 2005) and PINK1 knockdown (Yang et al., 2006), as a load-dependent mechanism predicts, although another PINK1 study found loss mainly in PAL, PPM and PPL2 with PPL1 least affected (Wang et al., 2006), and a systematic recount with a nuclear reporter found no loss at all in α-synuclein, PINK1 or parkin flies (Navarro et al., 2014). α-synuclein models are reported to lose a group of about 15 anterior-medial PAM neurons projecting to the β and β’ lobes (Riemensperger et al., 2013), which load does not predict, so if both reports hold the two pathologies follow different vulnerability rules. Within PPL1 the model predicts the order PPL101, PPL103, PPL107.

The order of loss sets the timing of the deficit, and passive readouts never show it. Removing PAM neurons in random order (five orders), highest-load first (the selective-vulnerability analogue) or lowest-load first changes when the appetitive memory of MBON09 fails. Under random loss the index is 0.91 ± 0.06 at 25 %, 0.86 ± 0.28 at 50 %, 0.67 ± 0.39 at 75 % and 0.11 ± 0.19 at 90 % (mean ± SD): reliable to 50 %, and at 75 % bimodal, intact in some orders and gone in others, according to how many of the compartment’s own 52 PAM neurons happen to survive. Highest-load-first loss keeps it at 0.87 at 75 %; lowest-load-first loss, which takes the compartment’s PAM12, PAM13 and PAM14 early, abolishes it (0.00) at 75 %. Under every order and at every fraction, including 100 %, the passive odour response of the MBON population (6.5 ± 0.2 Hz against 6.5 intact) and the locomotor descending neurons (left 20.6 ± 1.3 Hz against 20.0, right 9.8 ± 1.0 against 8.8; 70 lesioned runs) are unchanged. In this model a climbing-type readout cannot report dopaminergic loss; a conditioning assay reports it from 75 to 90 % loss. Drug screens that rely on climbing alone are, on this account, blind to the lesion they target.

Old memories are kept, new ones cannot be formed. A memory written before the loss survives removal of 50, 75 and 100 % of PAM neurons intact (MBON09 index 0.96 ± 0.06, 0.95 ± 0.07 and 1.00 ± 0.00 after the lesion, five orders each, against 1.00 before), while acquisition after the same loss is already unreliable at 75 % and fails at 90 %. The memory lives in the Kenyon-cell-to-MBON synapse and dopamine is only the writing signal, so retention and acquisition dissociate, as they do between the preserved habits and impaired learning of patients. Fly models can test this directly by training before and after the onset of degeneration.

Dopamine replacement does not restore learning and can corrupt it. Adding tonic extrasynaptic dopamine of 10 to 60 Hz-equivalent to every surviving dopaminergic neuron, the analogue of L-DOPA feeding, leaves paired learning at 1.0 whether or not 75 % of PAM is lost (the reserve already covers that loss). Its effect depends entirely on whether the gate adapts to the tonic level. If it does, nothing changes (unpaired depression at most 0.21). If it does not, odour alone now depresses the appetitive compartments without any pairing (MBON09 0.75 at 30 Hz, 1.00 at 60 Hz; MBON33 0.33) and leaves the others untouched: plasticity that has

lost its specificity, in the compartments that learn, the circuit form of the overdose and aberrant-plasticity accounts of L-DOPA side effects. The two cases differ in a measurable way, the odour response of reward-compartment MBONs in unpaired, L-DOPA-fed flies.

Compensation by survivors has a hard limit. Letting the surviving PAM neurons fire $(1 - f)^{-\alpha}$ times harder after a loss of fraction f (up to 10-fold at 90 %) adds nothing at 50 and 75 % loss (0.86 ± 0.32 and 0.66 ± 0.49 with full compensation against 0.81 ± 0.29 and 0.69 ± 0.42 without, five orders) and rescues nothing at 90 % (0.12 ± 0.18), where only 4 of the 52 PAM neurons of the MBON09 compartment survive: the gate reads the compartment's mean, so hyperactivity of the remainder can extend the reserve only while enough neurons remain in the compartment. The reserve is set per compartment by the number of surviving neurons and the Kenyon-cell drive, not by a global dopamine level.

These predictions do not depend on the fitted parameters. Repeating the key measurements with the reward gain, the gate constant κ and the plasticity rate η each halved and doubled, and with the bitter relay route removed (eight parameter sets), gives the same four qualitative outcomes in every set: MBON09 learns from sugar (index 0.56 to 1.00), learning survives 75 % PAM loss (0.53 to 1.00) and fails at 90 % (−0.07 to 0.12), a memory stored before a 75 % loss is retained (0.50 to 1.00, equal to the intact index of that set), and tonic dopamine on a non-adapted gate depresses the appetitive compartment without pairing (0.42 to 1.00, scaling with κ and η). Only magnitudes move with the parameters; the reserve threshold does not, because it is set by the number of surviving neurons in the compartment.

Further checks. The compartment map and the reserve are the same for a second odour (ORN DL5, DM2 and VM2: MBON07 1.00, MBON02 0.78, both intact after 90 % PAM loss). The reserve does depend on Kenyon-cell drive, in the direction the gate rule predicts: doubling the Kenyon-cell synapses onto MBON09 raises its odour response from 13 to 48 Hz, lowers its learning index from 0.88 to 0.37 and brings its failure forward (0.27 at 75 % loss, 0.08 at 90 %), while halving them silences it. Applying all three rules of Jo (2026) together (neuron loss, synapse loss, threshold shift) keeps the hub-shuffled-antihub ordering in both regimes and moves the antihub-to-hub ratio in the attractor regime to 2.1, nearer the reported 3.0. A different neuron model does not rescue the reference regime: spike-frequency adaptation (threshold rising 0.1 to 4 mV per spike, decaying over 1 s) removes the attractor but leaves no stable sparse odour code, the response decaying within the second at every strength and vanishing at the 1 mV used by Jo; with APL strengthened the negative result is unchanged (floor 1.93, PPL1 0.97, PAM 1.57, random 1.01), and the positive arm cannot be run because MBON09 no longer responds. On the male connectome at its usable operating point (scale 0.35, APL ×5) only three MBON types respond to the odour and none learns, so the positive arm is not testable there without a male-specific operating point.

Structural damage ranks correctly in both regimes

To test whether the attractor also compromises published lesion results, we re-ran the neuron-loss rule of Jo (2026) on the female connectome: each non-sensory neuron is removed with probability $0.5 \cdot s \cdot v$, where v is its percentile rank by synaptic contact count (hub map), the same values shuffled, or reversed (antihub map), at severity s = 0.5, 8 replicates per map, in both regimes.

Regime	Map	Neurons removed	Contacts lost	Odour-evoked spikes (healthy = 100 %)	KC fraction active	Phenotype distance (noise floor)
attractor	hub	12.2 %	33 %	52 % (IQR 50 to 59)	0.49	12.5 ± 1.3 (0.92)
attractor	shuffled	11.0 %	23 %	72 %	0.57	7.7 ± 1.6
attractor	antihub	9.9 %	12 %	90 %	0.63	3.6 ± 0.2
sparse	hub	12.2 %	33 %	65 % (IQR 62 to 94)	0.16	12.9 ± 7.8 (1.31)
sparse	shuffled	11.0 %	23 %	81 %	0.02	5.8 ± 0.8
sparse	antihub	9.9 %	12 %	97 %	0.03	1.9 ± 0.9

The ordering reported by Jo (hub worst, antihub mildest, damage tracking contacts lost) holds in both regimes, so it is not an artefact of the attractor. The antihub-to-hub ratio is 1.5 to 1.7 here against 3.0 in Jo, whose protocol also includes synapse loss and a threshold shift, drives projection neurons directly, and runs on the male connectome. The sparse regime adds a caveat: the two APL neurons rank at the top of the hub map (removal probability 0.25 at s = 0.5), and in the 3 of 8 replicates in which one was lost the brain flipped into the attractor, with the Kenyon-cell fraction rising from 0.01 to 0.41, MBON rates from 3 to 90 Hz and whole-brain activity to 108 % of healthy. A whole-brain spike count can therefore rise under damage, and its sign depends on the operating point. Jo's own need to add adaptation to end activity that persisted after an odour is, we suggest, the same attractor seen from the other side.

Embodiment, inverse inference, and the male connectome

Embodied readout. We coupled the brain model to NeuroMechFly v2 by mapping the mean firing rate of locomotor descending neuron types per side, relative to the intact brain under the same stimulus, onto the two-sided drive of a hybrid turning controller. The controller responds linearly to this drive (12.8 mm/s at drive 1.0, 6.5 at 0.5, 2.1 at 0.2) and climbs a 30 degree slope. Under single-modality stimulation no dopaminergic lesion changed descending output at all. Under combined odour and mechanosensory drive, complete PPL1 loss reduced walking and climbing speed by 13 % in the reference model, against 1 % and 5 % for size-matched random controls; Kenyon-cell loss reduced it by 32 %. With the modulator layer the same lesion left speed within control noise. The direction is wrong either way: fly Parkinson models show decline, and the attractor-dependent reference result is the only one with the right sign.

Inverse model. We trained a ridge surrogate and a neural posterior estimator on simulator pairs to infer lesions from phenotypes. On a dopamine-only sweep of 80 simulations, the posterior recovered PAM and PPL1 loss fractions with median errors of 0.15 and 0.11 against a prior baseline of 0.33, while PPL2 and PPM1 remained unidentifiable. On a subtype-resolved sweep of 240 simulations over 32 cell types, only the strongest labelled line was recovered. Enlarging that sweep to 1,000 simulations (run on an Apple M4 desktop in 65 minutes) raised the ridge surrogate's cross-validated R^2 from 0.33 to 0.52, yet the posterior's error on lesioned entries (0.41) still exceeds the constant prior-median baseline (0.26): six subtypes are recovered (PPL101 r = 0.91, PPL103 0.89, PPL105 0.87, PPL106 0.70, PAM08 0.67, PAM04 0.64) and the other 26 are not. The same 1,000-simulation sweep repeated in the sparse regime (APL ×5) gives a ridge R^2 of −0.02, a posterior error of 0.46 against the 0.26 baseline, a detection rate of 2 % at a 3 % false-alarm rate, and no subtype with r above 0.5: the six lines readable in the attractor vanish with it. With the reward route supplied, a third sweep of 300 random lesions used the learning readout itself as the phenotype (learning index per MBON after sugar and af-

ter bitter pairing, plus the odour response; APL ×3). The 32-dimensional inverse problem stays ill-posed (ridge R^2 −0.06, detection 0.33 against a false-alarm rate of 0.31), because only two or three MBONs learn and the phenotype carries about that many informative dimensions, but the signal that exists is the right one: PAM12, the most numerous PAM type of the MBON09 compartment, is read from the learning phenotype with $r = 0.56$, against 0.13 and −0.44 from the passive phenotypes, together with PPL107 (0.79), PPL101 (0.53) and PPL106 (0.51). The machinery works; what it recovers is limited by how much of the lesion the phenotype encodes, which for passive activity is the attractor and nothing else, and for learning is the dopaminergic input of the compartments that learn. The inverse problem that is well posed is therefore one subtype per learning compartment, which the reserve curves below answer directly.

Male connectome. We built the same pipeline on MaleCNS v1.0 from the released flat files (164,606 traced neurons, 25.6 million edges) and carried the stimulus battery across by cell type. At the published alignment scale of 0.581 the male model enters the attractor under every stimulus and its trial-to-trial noise floor rises to 20.4, against 1.68 for the female. Reducing the synaptic scale to 0.3 lowers the floor to 3.25, but then no PAM neuron fires at all and every lesion effect sits below noise. Strengthening APL inhibition 5-fold at the published scale, as for the female, lowers the male floor to 10.4, and in that regime complete loss of PPL1 (distance 9.0) or PAM (1.2) is again indistinguishable from size-matched random loss (RND16 4.4, RND307 9.9), as in the female. Scanning the synaptic scale with APL strengthened gives the male a usable operating point at 0.35 (floor 1.81, the female's 1.91), where loss of 80 to 100 % of PPL1 moves the phenotype by 2.7, 1.5 times the floor and above random loss of 307 neurons (1.1), while PAM is silent at that scale and its loss has no effect by construction. We therefore report no clean sex comparison: the female gives a negative answer in its sparse regime, and in the male a small PPL1 effect is readable at one operating point while PAM is not, so the two connectomes could not be placed at comparable points under this model, which is a result about the model rather than about the brains.

Discussion

Connectome-constrained integrate-and-fire models predict sensorimotor responses well, and that success has encouraged their use for questions further from sensory transformation. Our results mark one boundary of that range. The model does not route reward or punishment to dopaminergic neurons, so it cannot represent the teaching signal of associative memory, and it cannot relate dopaminergic cell loss to behaviour in the way fly models of Parkinson's disease do.

The boundary moves once the missing signal is supplied explicitly. A single extrasynaptic reward term, with no other change, makes the wiring predict which compartments store appetitive memory, and the prediction agrees with behavioural assignments for every odour-responsive compartment they cover. The same model then speaks to the questions fly Parkinson models ask, and three of its answers are not obvious in advance: passive and locomotor readouts stay blind to dopaminergic loss that a conditioning assay detects, because the loss removes a writing signal rather than a drive; the reserve before a memory fails is set per compartment by surviving neuron number and Kenyon-cell drive, so the order in which subtypes die decides which deficit appears first; and dopamine replacement cannot restore a writing signal it does not time, so its only effect is on specificity. Each is a claim about circuits that cell-type-specific ablation, imaging of PPL1 during pairing, and L-DOPA feeding in flies can confirm or refute, and the connectome-derived vulnerability ranking (PPL1 twelvefold above PAM in synaptic load) matches the PPL1 loss reported for parkin and PINK1 flies, with the caveat that the fly literature disagrees on whether, and where, dopaminergic neurons are lost at all.

The finding also shows how a model can produce a confident wrong answer. Taken at the default operating point, our lesion experiment yields an attractive result: loss of 16 PPL1 neurons moves the phenotype six times more than loss of 16 random neurons, through three identifiable labelled lines, and an inverse model recovers

those lines from the phenotype alone. Every step is internally consistent. It is also entirely a product of an odour-evoked attractor in which three quarters of Kenyon cells fire and output neurons saturate, and it vanishes when Kenyon-cell activity is brought to a realistic level. Reporting a dopaminergic lesion effect from such a model therefore requires showing where the model sits relative to that attractor.

What the result does not show is that connectomes are the wrong substrate. For punishment the wiring suffices: 20 relay neurons carry bitter taste to PPL1 once their synapses are 4-fold stronger than the single global weight allows, and the model names them. For reward the synaptic wiring from sugar to PAM is not merely weak but missing: the octopaminergic neurons that carry it in vivo receive no sugar input and barely contact PAM, so the teaching signal must travel a route the connectome does not record, extrasynaptic octopamine, and a model that wants it has to add that term explicitly. Likely contributors in both cases are the absence of neuromodulation and adaptation, and of any tuning of synaptic strength beyond one constant, together with threshold settings that let a strong recurrent population latch while sparse polysynaptic relays stay subthreshold. Structural questions fare better: the ranking of damage by connectivity (Jo, 2026) survives the change of regime, although a whole-brain activity readout can change sign when inhibitory hubs are lost.

Three directions follow. First, calibrate the operating point: fit the global synaptic scale and inhibitory gain so that odour responses reproduce measured Kenyon-cell sparsity before any lesion study. Second, test conduction directly: identify which relay neurons on the sugar-to-PAM path fail to reach threshold and what changes recruit them. Third, if the teaching signal must be injected rather than evoked, say so explicitly and treat the resulting model as a model of plasticity rather than of reinforcement.

Our own study inherits limitations. The sparse regime was produced by scaling one inhibitory connection fivefold, a manipulation chosen for its effect rather than measured; it should be treated as a sensitivity analysis. Lesions are instantaneous while the experimental phenotypes develop over 10 to 30 days, so compensation is absent. The mapping from descending neurons to walking speed is an assumption, not a fitted model. The compartment identities we use are connectivity-based, and we have not asserted their correspondence to the classical Greek compartment names.

Methods

Connectome data. Female: FlyWire v783 connectivity and completeness tables as distributed with the reference model, 138,639 neurons and 15,091,983 signed connections, with cell-type, class, side and neurotransmitter annotations from the FlyWire annotation release. Male: MaleCNS v1.0 flat connectome files (body annotations, neurotransmitters, segment-to-segment weights), restricted to traced neurons with a superclass, giving 164,606 neurons and 25,558,671 edges. Male superclass labels were mapped to the female vocabulary; acetylcholine, dopamine, octopamine and serotonin were treated as excitatory, GABA, glutamate and histamine as inhibitory.

Network model. Leaky integrate-and-fire as in the reference model, whose constants come from Kakaria and de Bivort (2017; membrane), Jürgensen et al. (2021; synaptic time constant), Lazar et al. (2021; refractory period) and Paul et al. (2015; delay): resting and reset potential -52 mV, threshold -45 mV, membrane time constant 20 ms, synaptic time constant 5 ms, refractory period 2.2 ms, synaptic delay 1.8 ms, 0.275 mV per synapse, 0.1 ms timestep. Stimuli are Poisson inputs onto the target neurons with weight 0.275 x 250 mV and no refractory period, as in the reference. Our implementation updates only neurons away from rest, which is equivalent to the dense form up to a 1e-3 mV sleep threshold and about five times faster; a dense implementation and a torch implementation are included and tested for equivalence.

Lesions. A lesion is a vector of loss fractions over named groups. Neurons within a group are ordered once per seed so that larger fractions are supersets of smaller ones. Removal zeroes all incoming and outgoing synapses;

silencing zeroes outgoing only; scaling multiplies them. A pattern-preserving variant keeps zeroed entries explicit, required by the modulation layer.

Stimulus battery and phenotype. Six stimuli: sugar (32 gustatory neurons), water (30), bitter (42), odour ORN_DM1 (68), wind and gravity (481) and head bristles (305), at 150 Hz except head bristles at 100 Hz. The phenotype is the mean log firing rate, 100 to 500 ms, of 109 readout populations: descending-neuron cell types with at least four neurons, pooled remainder, and output neuron types. Distances are L2 in this space against the mean intact phenotype; the noise floor is the mean pairwise distance between three intact repeats run with different simulation seeds.

Modulation layer. Compartments are pairs of a dopaminergic neuron and an output neuron connected directly with at least 2 mV. Fast dopaminergic synapses onto Kenyon cells and output neurons are removed. Every 20 ms the Kenyon-cell synapses onto each output neuron are multiplied by exp(-kappa (rbar - b)), clipped to [0, 3], where rbar is the compartment's dopaminergic rate low-passed with a 200 ms time constant and b its tonic value measured in the intact brain with the gain switched off. With plasticity enabled the baseline weights are depressed in proportion to the product of Kenyon-cell rate and the phasic excess of dopamine. Parameters were fitted by grid search over simulated conditioning.

Embodiment. NeuroMechFly v2 (flygym 2.1.1) with the hybrid turning controller and preprogrammed steps, flat ground and a gravity-tilted variant, 0.1 ms physics timestep, 1 s per condition. Drive per side is the ratio of lesioned to intact mean rate of the locomotor descending types, clipped to [0, 1.5].

Inference. Lesion vectors were drawn from a sparse prior: each group lesioned with probability 0.5 (mixed and dopamine-only sweeps) or 0.15 (subtype sweep), with loss fractions uniform on [0, 1]. Surrogates: ridge regression and an ensemble of five multilayer perceptrons, selected by five-fold cross-validated R-squared. Posteriors: neural posterior estimation with a neural spline flow, sampled without prior rejection and clipped to the unit box because rejection acceptance falls below 1 % in 32 dimensions. Recovery is reported on held-out simulations over lesioned entries only, against a predict-zero and a prior-median baseline.

Reinforcer pathway. In the sparse regime the reinforcer set (sugar 32, bitter 42 or water 30 gustatory neurons) was driven at 150, 300, 600 and 1000 Hz for 500 ms; the active set is every neuron above 5 Hz after 100 ms. Hop analysis summed the signed synaptic input (mV) from the gustatory set onto each neuron and, for the active set, the rate-weighted input (mV per 10 ms) onto every neuron that projects to the target cluster. Relays are the 20 neurons outside the target cluster with the largest product of rate-weighted input from the active set and total output onto the target; the input hop (active set to relays), the output hop (relays to target) or both were multiplied by k in {1, 2, 4, 8, 16} and the target cluster rate measured at 150 Hz. Conditioning with the restored pathway used the modulation layer with the fitted parameters, a 1 s pairing of odour with bitter at 150 Hz each, 400 ms rest, and odour tests of 500 ms before and after.

Neuron-loss re-test. Following Jo (2026), the vulnerability of each neuron is its percentile rank by synaptic contact count (in plus out, contacts = synaptic weight divided by the unit weight), shuffled across neurons for the shuffled map and reversed for the antihub map; every non-sensory neuron is removed with probability $0.5 \cdot s \cdot v$ at $s = 0.5$, independently per replicate (8 replicates per map, 3 healthy repeats), in the attractor (APL ×1) and sparse (APL ×5) regimes. Readouts: spikes of all neurons during 100 to 500 ms of ORN DM1 odour at 150 Hz, fraction of Kenyon cells above 5 Hz, mean MBON rate, and the phenotype vector of the 109 readout groups under that odour. Synapse loss and the threshold shift of the original protocol were not applied.

Teaching-signal routes and Parkinson protocols. The reward term adds 0.5 × the mean rate of the 32 sugar gustatory neurons to the effective dopamine rate of every surviving PAM neuron in the gain and plasticity rules (ModulatedNetwork.add_extrasynaptic); the bitter route is the 4-fold relay strengthening above. Compartment map and reserve: odour test 500 ms, pairing 1 s, rest 400 ms, test 500 ms, learning index (pre − post)/pre, at

APL ×5 and ×3, lesions of PAM and PPL1 at 0 to 100 % (two orders) and 307 random neurons; unpaired noise from the same protocol with odour alone (3 seeds). Loss order: PAM neurons removed in random order, or sorted by synaptic load (in plus out contacts) descending or ascending. Retention: pairing in the intact network, then rows and columns of the removed neurons zeroed in the trained network (stored Kenyon-cell-to-MBON weights kept), then the odour test. Tonic dopamine: a constant added to the effective rate of every surviving PAM and PPL1 neuron, with the baseline measured either with it (adapted gate) or without (non-adapted). Compensation: the effective rate of surviving PAM neurons multiplied by $(1 - f)^{-\alpha}$, $\alpha \in \{0, 0.5, 1\}$. Vulnerability: contacts per neuron from the synaptic weight divided by the unit weight.

Reproducibility. All experiments are scripted and checkpointed. Simulation seeds and lesion orders are reported separately; curves are averaged over 3 to 5 lesion orders and 1 to 2 simulation seeds as stated per figure.

Data, code and figures

All code, result tables and figures are in the repository github.com/kaantiftikci/connectome-lif-lesions (MIT licence; citation metadata in CITATION.cff). The version used for this manuscript is release v1.0.0, archived at Zenodo under doi:10.5281/zenodo.23808330. Connectome data are public: FlyWire v783 tables are redistributed with the reference model repository, FlyWire annotations come from the flyconnectome annotation release, and MaleCNS v1.0 flat files are served from the project's public storage. No new experimental data were produced.

Figures

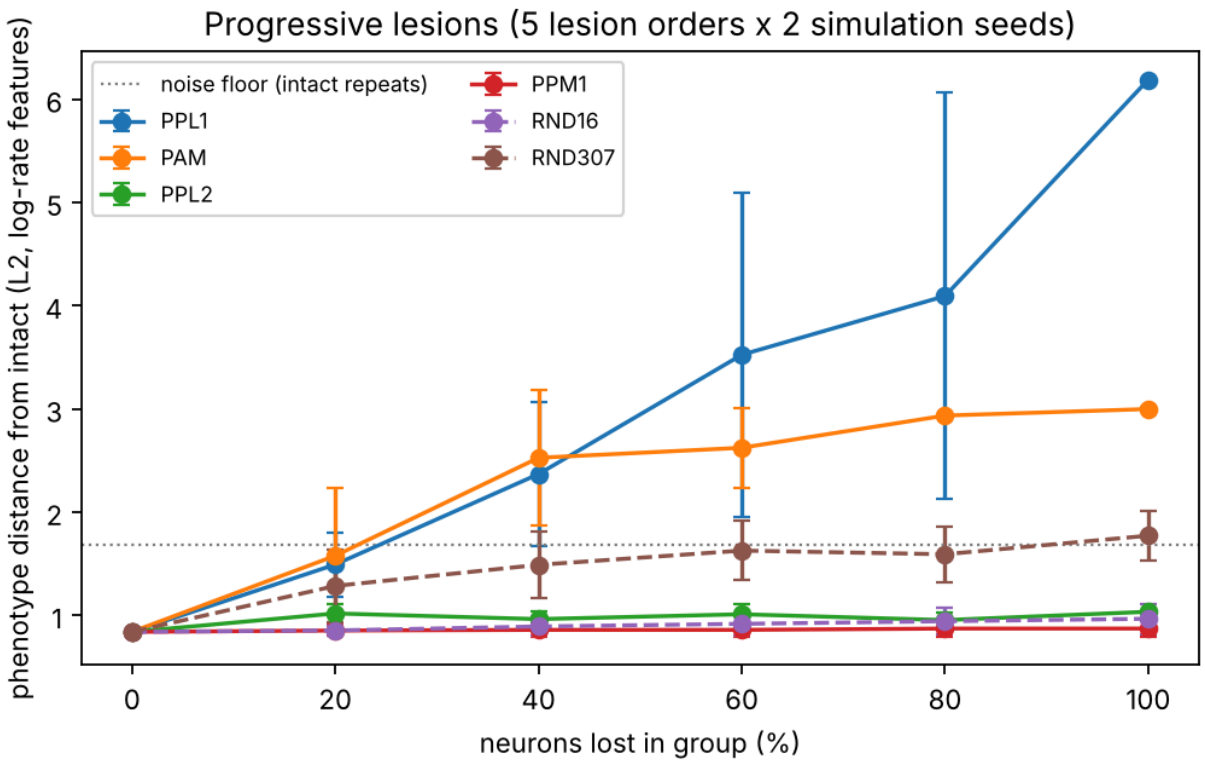


Figure 1. Progressive lesion curves in the attractor regime.

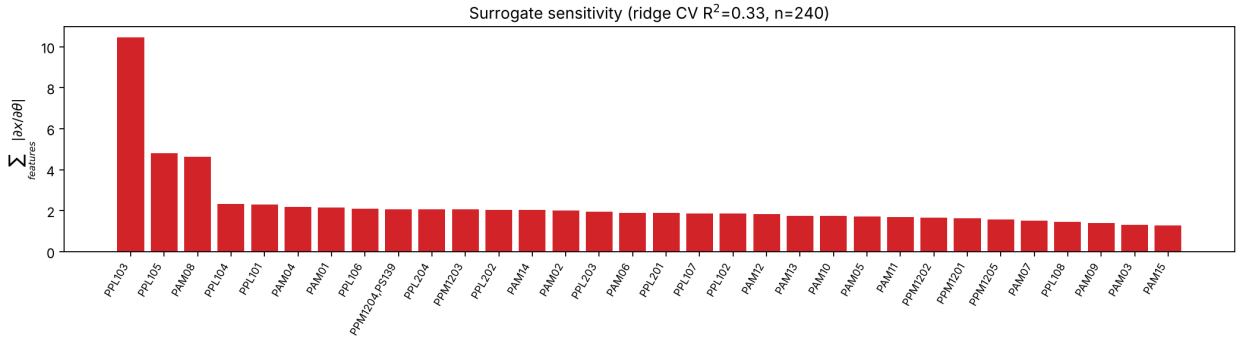
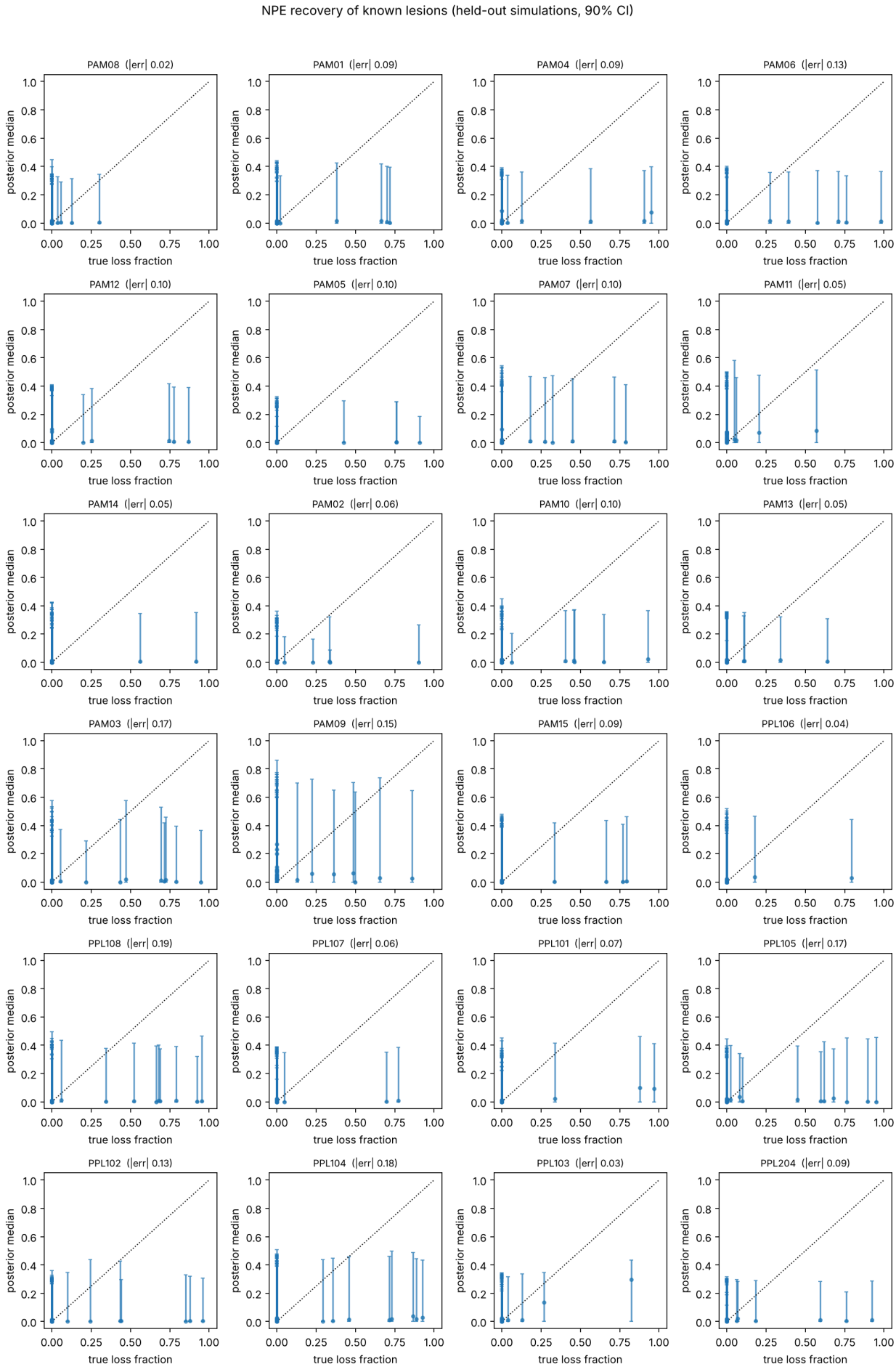


Figure 2. Surrogate sensitivity per dopaminergic subtype.


NPE recovery of known lesions (held-out simulations, 90% CI)
PAM08 (|err| 0.02)
PAM01 (|err| 0.09)
PAM04 (|err| 0.09)
PAM06 (|err| 0.13)
PAM12 (|err| 0.10)
PAM05 (|err| 0.10)
PAM07 (|err| 0.10)
PAM11 (|err| 0.05)
PAM14 (|err| 0.05)
PAM02 (|err| 0.06)
PAM10 (|err| 0.10)
PAM13 (|err| 0.05)
PAM03 (|err| 0.17)
PAM09 (|err| 0.15)
PAM15 (|err| 0.09)
PPL106 (|err| 0.04)
PPL108 (|err| 0.19)
PPL107 (|err| 0.06)
PPL101 (|err| 0.07)
PPL105 (|err| 0.17)
PPL102 (|err| 0.13)
PPL104 (|err| 0.18)
PPL103 (|err| 0.03)
PPL204 (|err| 0.09)
posterior median
true loss fraction
0.0
0.2
0.4
0.6
0.8
1.0
0.00
0.25
0.50
0.75
1.00

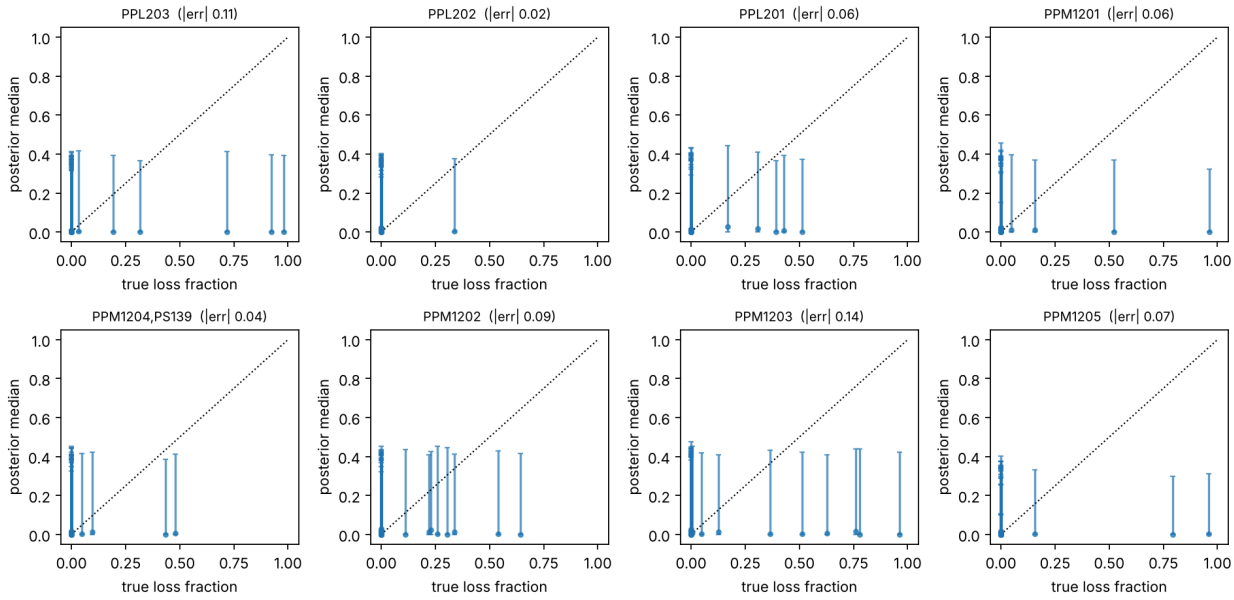


Figure 3. Posterior recovery of known lesions.

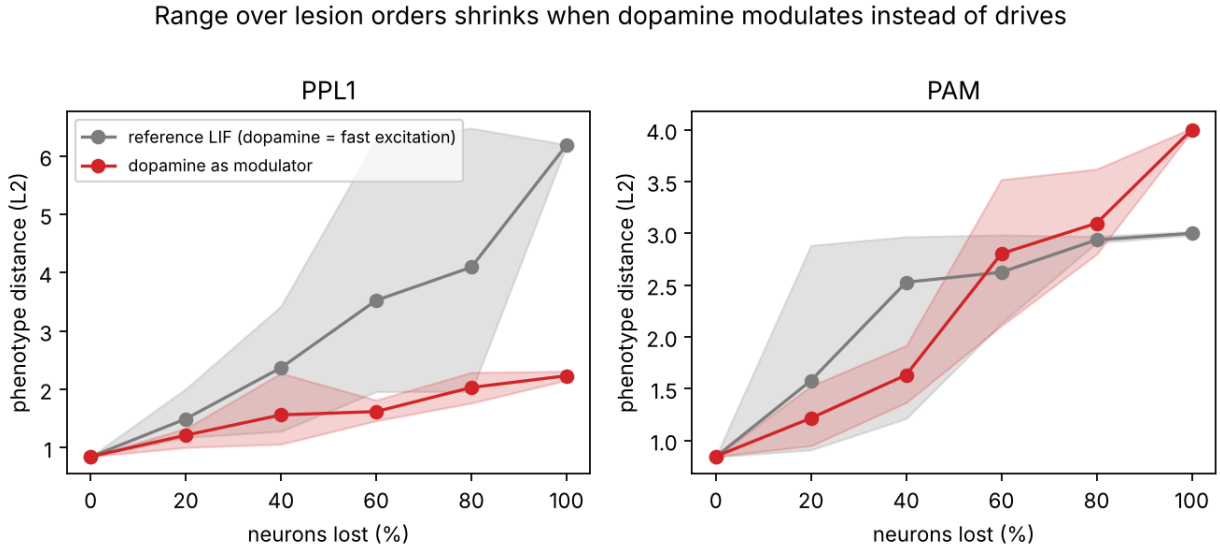


Figure 4. Reference model against the dopamine-as-modulator model.

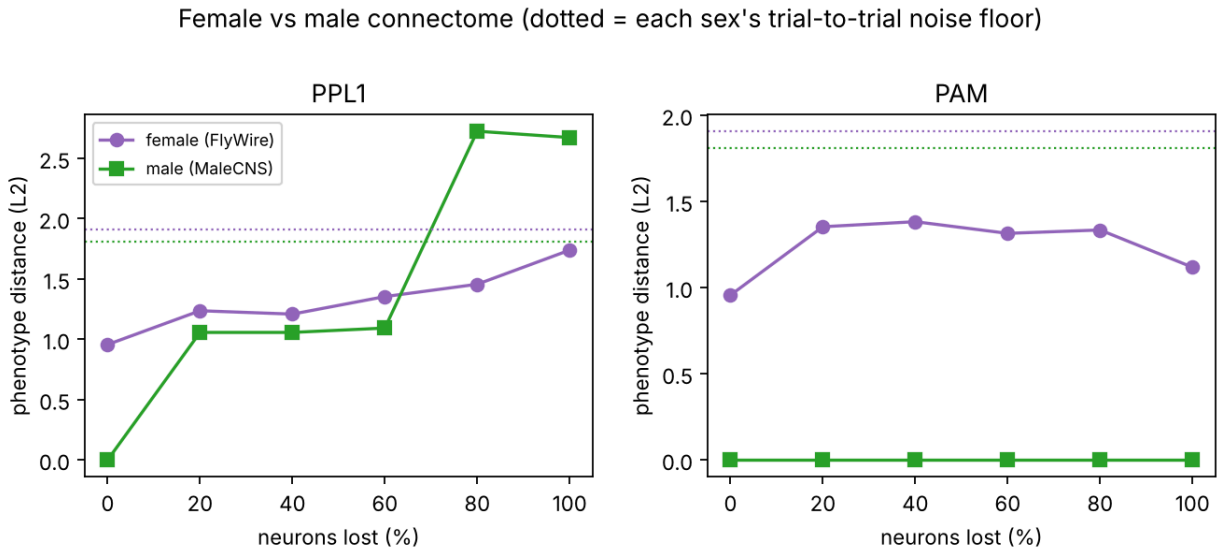


Figure 5. Female (sparse regime) and male (scale 0.35, APL ×5) lesion distances.

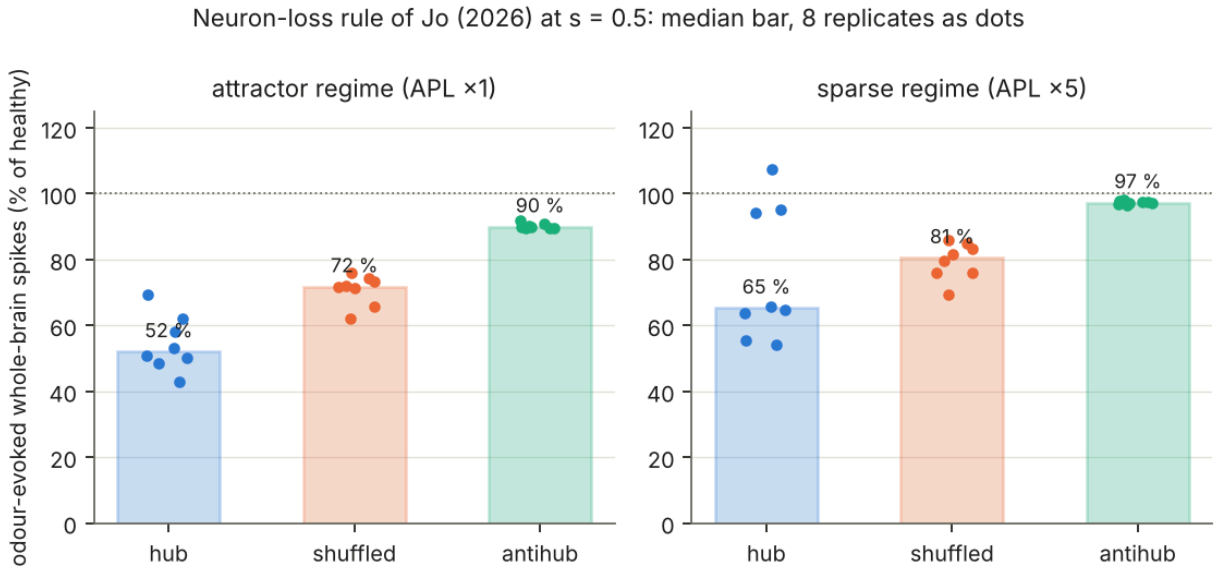


Figure 6. Neuron-loss rule of Jo (2026) in both regimes.

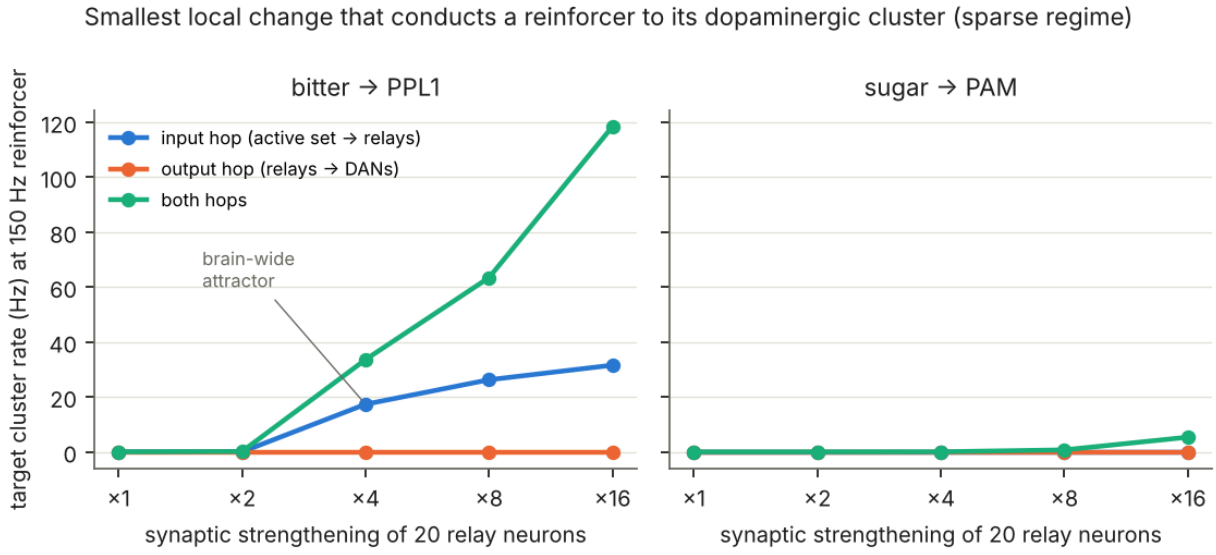


Figure 7. Smallest local change that conducts a reinforcer to its dopaminergic cluster.

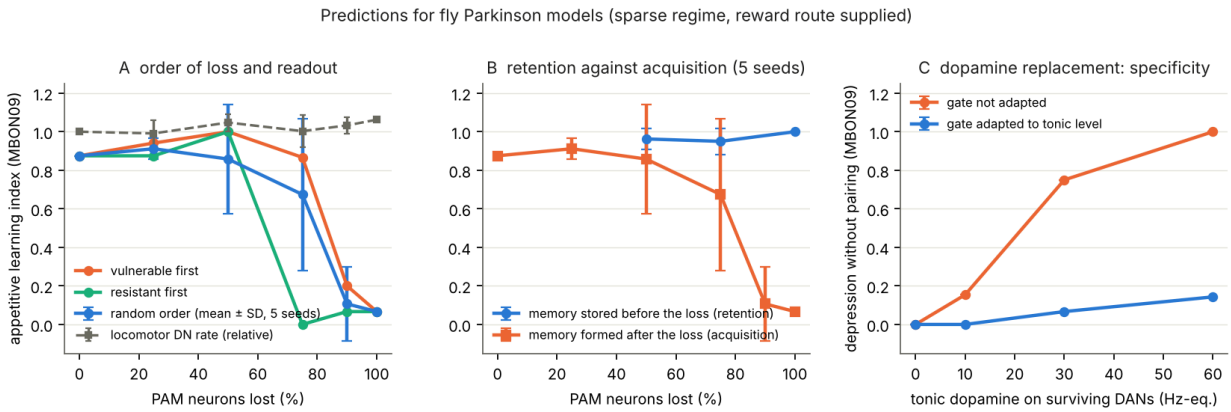


Figure 8. Predictions for fly Parkinson models: order of loss and readout (mean ± SD over five random orders), retention against acquisition, dopamine replacement and specificity.

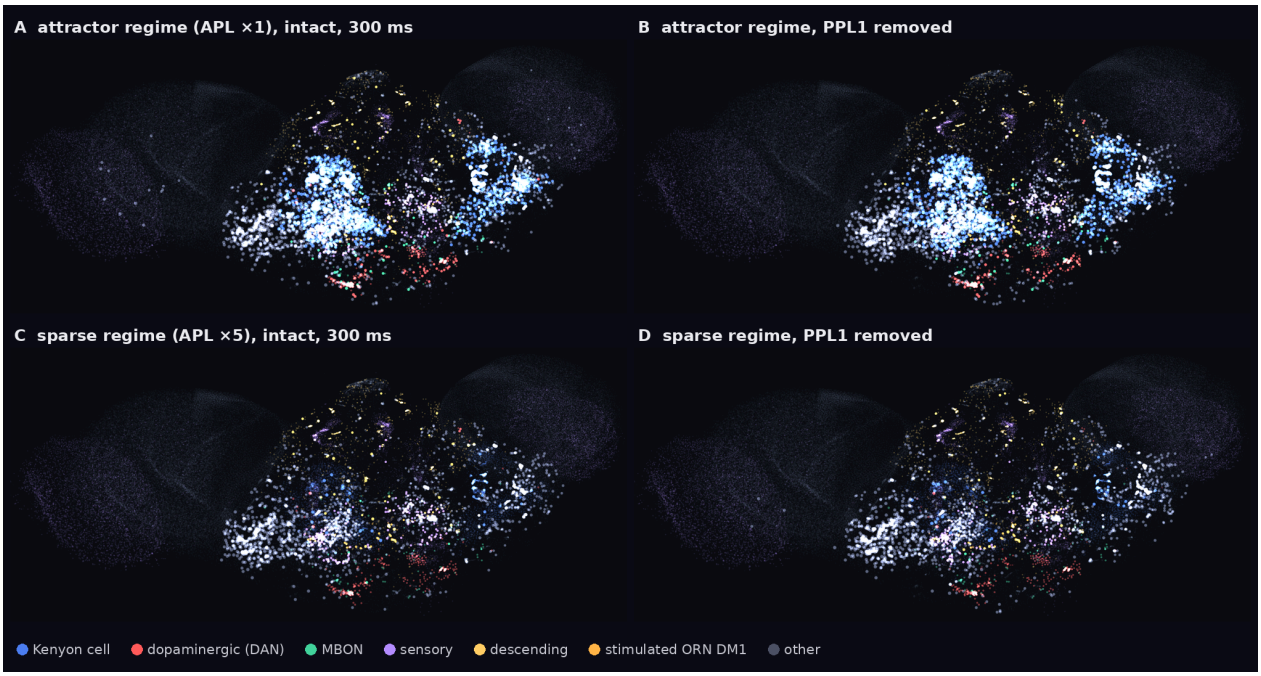


Figure 9. Whole-brain spiking activity at 300 ms of a 150 Hz ORN DM1 odour, rendered from the interactive viewer (viewer/ in the repository): (A) attractor regime, intact; (B) attractor regime, PPL1 removed; (C) sparse regime (APL ×5), intact; (D) sparse regime, PPL1 removed. Points are neurons at their FlyWire soma positions, coloured by class and lit by recent spikes; in the attractor regime three quarters of the Kenyon cells (blue) are firing, in the sparse regime the odour code is confined to a few percent of them.

Figure legends. Figure 1. Progressive lesion curves in the attractor regime: phenotype distance from the intact brain against fraction of neurons lost for PPL1, PAM, PPL2, PPM1 and random central controls of 16 and 307 neurons; mean and SD over 5 lesion orders and 2 simulation seeds; dotted line, noise floor between intact repeats. Figure 2. Surrogate sensitivity, summed absolute gradient of the phenotype with respect to loss fraction, for 32 dopaminergic subtypes (240 simulations). Figure 3. Posterior recovery of known lesions on held-out simulations: true against posterior-median loss fraction with 90 % interval, one panel per subtype. Figure 4. Reference model against dopamine-as-modulator model for PPL1 and PAM: mean distance with the range over lesion orders shaded. Figure 5. Female (sparse regime) and male (synaptic scale 0.35, APL ×5) lesion distances for PPL1 and PAM with each sex's noise floor marked; the published male scale of 0.581 is not plotted because its floor (10.4 with APL ×5, 20.4 without) exceeds every lesion effect. Figure 6. Neuron-loss rule of Jo (2026) at severity 0.5 under the hub, shuffled and antihub vulnerability maps, in the attractor and sparse regimes: odour-evoked whole-brain spikes as a percentage of the healthy brain, median bar and 8 replicates as dots; the sparse-regime hub replicates above 100 % are those in which an APL neuron was removed. Figure 7. Target dopaminergic cluster rate at a 150 Hz reinforcer when the input hop, the output hop or both hops of the 20 strongest relay neurons are strengthened 1- to 16-fold, for bitter to PPL1 and sugar to PAM; the 4-fold input-hop point for bitter is a brain-wide attractor. Figure 8. Predictions for fly Parkinson models: (A) appetitive learning index of MBON09 against PAM loss in random, highest-load-first and lowest-load-first order, with the relative locomotor descending-neuron rate; (B) a memory stored before the loss against one formed after it; (C) depression of MBON09 without any pairing under tonic dopamine on surviving neurons, with the gate adapted or not adapted to the tonic level. Figure 9. Whole-brain spiking activity at 300 ms of a 150 Hz ORN DM1 odour, rendered from the interactive viewer (viewer/ in the repository): (A) attractor regime, intact; (B) attractor regime, PPL1 removed; (C) sparse regime (APL ×5), intact; (D) sparse regime, PPL1 removed. Points are

neurons at their FlyWire soma positions, coloured by class and lit by recent spikes; in the attractor regime three quarters of the Kenyon cells (blue) are firing, in the sparse regime the odour code is confined to a few percent of them.

Author contributions, AI assistance and competing interests

K.T. conceived the study, defined the research questions, directed the work, reviewed every design decision and is responsible for the content. All code, simulations, analyses and the first draft of this manuscript were produced with an AI assistant based on a large language model, working in an agentic coding environment under K.T.'s direction over two days; part of the compute ran on K.T.'s own machines. The assistant is not an author, and its role is disclosed in full here, in line with publisher guidance on AI-assisted research. The author declares no competing interests and received no funding for this work.

Where the AI assistant was used. (1) Study design: the assistant proposed the lesion protocol, the sparse-regime control, the reinforcer-path analysis, the teaching-signal arm and the Parkinson protocols; K.T. chose which proposals to pursue, set the research questions (the sex comparison, the positive arm, the Parkinson section, the parameter-sensitivity checks) and rejected or redirected others. (2) Software: the assistant wrote the simulator, the lesion, modulation, embodiment and inference code, the experiment scripts and the tests, and verified the simulator against the reference Brian2 implementation. (3) Simulations and analysis: the assistant ran the experiments, queued the sweeps and the CPU runs on K.T.'s machines, and computed every number reported here from the result files; K.T. supervised the runs and reported their outputs back. (4) Literature: the assistant drafted the comparisons with published compartment assignments and fly Parkinson models and, after K.T. opened network access, re-checked each cited claim against the primary source where it was reachable; the Navarro (2014) and Wang (2006) caveats came from that check. (5) Writing: the assistant wrote the first draft, the figures and the Turkish translation; K.T. read the text, asked for restructuring (the reframing around what the model can and cannot say, the positive arm, the Parkinson predictions) and approved the final content. (6) Not used: the assistant did not generate, alter or select data; every number comes from a scripted, checkpointed run in the public repository, and no experimental data were produced.

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